

Hawkeye EQMS · Change Control · Feature Guide

Pharma-EQMS comparison · click-by-click guide · personas + lifecycle walkthrough · AI map · test results.

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Tenant under test	Novex Pharma Inc. · 69e64e7869b2ba745d40bb89
Live frontend	https://hawkeye-frontend-dev-chi.vercel.app
Live backend	https://hawkeye-backend-dev.vercel.app

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1. Module overview

Module	Change Control
Pharma purpose	Control + document any proposed change to an approved process, material, equipment, specification, or procedure. Drive impact assessment, multi-level approval, implementation, effectiveness verification.
Compliance basis	ICH Q10 §3.2.3 (change management) · 21 CFR 211.100 · 21 CFR 820.30 (medical device) · EU GMP Annex 15
Hawkeye routes	/change-controls (frontend) · /api/universal/change-controls (backend)
Hawkeye model file	backend/src/models/ChangeControlModel.js
Tenant module flag	modules.CHANGE_CONTROL

Changes follow DRAFT → SUBMITTED → IMPACT_ASSESSMENT → UNDER_REVIEW (multi-step approval) → APPROVED → IMPLEMENTATION → VERIFICATION → CLOSED (or REJECTED / CANCELLED). The Wave 2 Regulatory Impact Classifier suggests US (CBE-30 / PAS-NDA) + EU (Type IA/IB/II Variation) routing from the change description.

2. Standard pharma EQMS vs Hawkeye — gap analysis

Each row is a capability a pharma QA team would expect from an EQMS deviation module. Hawkeye column shows what's actually shipped. **MET** = parity, **PARTIAL** = ships but with caveats, **GAP** = not implemented yet.

Pharma expectation	Hawkeye implementation	Status	Note
Change types (PROCESS / MATERIAL / EQUIPMENT / DOCUMENTATION / FACILITY / SUPPLIER / SPECIFICATION / OTHER) ICH Q10	changeType enum.	MET	
Risk level (LOW / MEDIUM / HIGH / CRITICAL) ICH Q9	riskLevel enum.	MET	
10-state lifecycle with role gates 21 CFR 211.100(b)	DRAFT / SUBMITTED / IMPACT_ASSESSMENT / UNDER_REVIEW / APPROVED / IMPLEMENTATION / VERIFICATION / CLOSED / REJECTED / CANCELLED.	MET	
Multi-step approval workflow with role + decision per step 21 CFR 211.100(b)	approvalSteps[] subdocument with role + userId + decision (PENDING/APPROVED/REJECTED/ABSTAINED). All APPROVED → status=APPROVED; any REJECTED → status=REJECTED.	MET	
AI regulatory impact classifier FDA AI guidance Jan 2025	POST /api/ai/change-control/classify-impact returns US (CBE-30 / PAS-NDA) + EU (Type IA/IB/II Variation) routing with reasoning.	MET	
Effectiveness verification after implementation ICH Q10 §3.2.3(d)	POST /:id/verify-effectiveness with verificationNotes + effectivenessCheck + effective boolean. effective=true → status=CLOSED; effective=false → bumps back to IMPLEMENTATION.	MET	
Linked records (CAPA / deviation / document / training) ICH Q10 §3.2.5	linkedDocumentIds, linkedCAPAIds, linkedTrainingIds arrays.	MET	
Affected products + markets capture FDA / EMA	affectedProducts[] + affectedMarkets[] arrays. Drives the AI impact classifier.	MET	
21 CFR Part 11 e-signature on each approval step 21 CFR Part 11 §11.50	approvalSteps[].signature field exists; not enforced today.	PARTIAL	Wire e-sig requirement
Per-tenant numbering (CCR-YYYY-NNNN) GMP traceability	Pre-save hook generates CCR-YYYY-NNNN. Bug fixed in v2.1: index now (tenantId, changeNumber) compound + per-tenant max-prefix scan.	MET	

3. Personas + role mapping

Login password for every persona: `EqmsDemo@2026` . Roles drive which API endpoints respond — see Hawkeye `roleMiddleware.permit()` .

Persona	Email	Responsibilities in this module	Lifecycle states they touch
Marcus Brown Regulatory Affairs (admin · originator)	<code>regulatory@novex-pharma.demo</code>	Drafts changes, fires AI impact classifier, drives the workflow.	DRAFT SUBMITTED IMPACT_ASSESSMENT
James Thompson Head of QA (admin · approver)	<code>qa.head@novex-pharma.demo</code>	Default approver on QA-impact changes.	UNDER_REVIEW
Sarah O'Brien Doc Control (admin · implementer)	<code>doc.control@novex-pharma.demo</code>	Implements doc-related changes (publishes new SOP rev linked to the change).	IMPLEMENTATION
Elena Vasquez VP Quality (tenant_admin · final approver)	<code>vp.quality@novex-pharma.demo</code>	Final approval on CRITICAL changes; signs effectiveness verification.	VERIFICATION CLOSED

4. Feature catalogue (click-by-click)

4.1 Change-control register

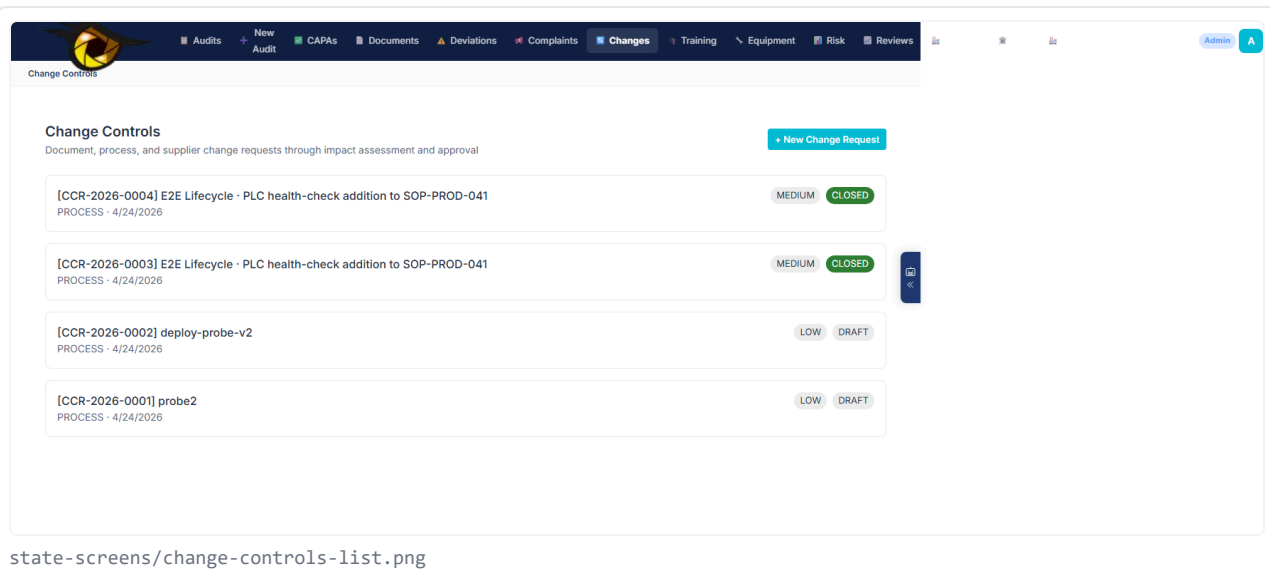
What it does	Lists every change with risk-level + status chips.
Menu / route	<code>/change-controls</code>
Available to roles	any viewer
Backend API	GET <code>/api/universal/change-controls</code>

CLICK-BY-CLICK

- NAVIGATE

 Click 'Changes' in the top nav
→ Page renders
- WAIT

 Spinner clears
→ Rows visible (empty if no changes seeded)



4.2 + New Change Request dialog

What it does	Author a new change request.
Menu / route	/change-controls · top-right '+ New Change Request' button
Available to roles	<code>createRoles</code>
Backend API	POST /api/universal/change-controls

CLICK-BY-CLICK

- CLICK** Click '+ New Change Request'
→ Dialog opens
- TYPE** Fill title (1 sentence)
→ required
- TYPE** Fill description (full narrative)
→ required
- CLICK** Pick changeType (PROCESS / MATERIAL / EQUIPMENT / DOCUMENTATION / FACILITY / SUPPLIER / SPECIFICATION / OTHER)
→ required
- CLICK** Pick riskLevel (LOW / MEDIUM / HIGH / CRITICAL)
→ required
- TYPE** (Optional) Fill affectedProducts[] + affectedMarkets[]
→ drives the AI classifier
- CLICK** (Optional) Add approvalSteps[] now or after submit
→ stepOrder + role + (userId)
- CLICK** Click 'Save'
→ Row appears with status=DRAFT; changeNumber=CCR-YYYY-NNNN auto-generated

FIELDS CAPTURED

Field	Required	Validation / values	Notes
title	Yes	string	
description	Yes	multi-line	
changeType	Yes	PROCESS MATERIAL EQUIPMENT DOCUMENTATION FACILITY SUPPLIER SPECIFICATION OTHER	
riskLevel	Yes	LOW MEDIUM HIGH CRITICAL	
affectedProducts	No	string[]	
affectedMarkets	No	string[] e.g. ['US', 'EU']	
approvalSteps	No	[[stepOrder, role, userId, decision: 'PENDING']]	

4.3 AI · Classify regulatory impact (Wave 2)

What it does	Given the change description + type + risk + products + markets, AI classifies US (CBE-30 / PAS-NDA / Annual) + EU (Type IA / IB / II Variation) routing with reasoning.
Menu / route	Change detail · 'Classify regulatory impact' button
Available to roles	regulatory · admin
Backend API	POST /api/ai/change-control/classify-impact
AI assistance	AI Free Gemini 2.5 Flash-Lite with regulatory reasoning prompt

CLICK-BY-CLICK

- CLICK** Click 'Classify regulatory impact'
→ Drawer opens
- CLICK** Click 'Generate'
→ After ~3 s: US route + EU route + rationale rendered. Auto-populates regulatoryImpact field

4.4 Submit (DRAFT → SUBMITTED)

What it does	Move the draft into the workflow.
Menu / route	Change detail · 'Submit' button (or PUT)
Available to roles	createRoles
Backend API	PUT /api/universal/change-controls/:id (status=SUBMITTED)

CLICK-BY-CLICK

- CLICK** Click 'Submit'
→ Status flips to SUBMITTED

4.5 Approve / reject step

What it does	Approver acts on a pending step. All APPROVED → status=APPROVED; any REJECTED → status=REJECTED.
Menu / route	Change detail · 'Approve' button (visible to assigned approver)
Available to roles	auditor admin tenant_admin reviewer workflow_manager
Backend API	POST /api/universal/change-controls/:id/approval

CLICK-BY-CLICK

1. **CLICK** Click 'Approve' on the row
→ Drawer with stepOrder + decision picker
2. **CLICK** Pick APPROVED / REJECTED / ABSTAINED
→ required
3. **TYPE** Fill comments (recommended)
→ stored on the step
4. **CLICK** Submit
→ Step decision recorded

4.6 Verify effectiveness (IMPLEMENTATION/VERIFICATION → CLOSED)

What it does	After the change is implemented, verify it achieved its intent.
Menu / route	Change detail · 'Verify Effectiveness' button
Available to roles	createRoles
Backend API	POST /api/universal/change-controls/:id/verify-effectiveness

CLICK-BY-CLICK

1. **CLICK** Click 'Verify Effectiveness'
→ Drawer with verificationNotes + effectivenessCheck + effective boolean
2. **TYPE** Fill verificationNotes
→ required
3. **TYPE** Fill effectivenessCheck (the criterion)
→ required
4. **CLICK** Tick effective=true (or false)
→ true → status=CLOSED; false → status=IMPLEMENTATION (re-work)
5. **CLICK** Submit
→ Status updates

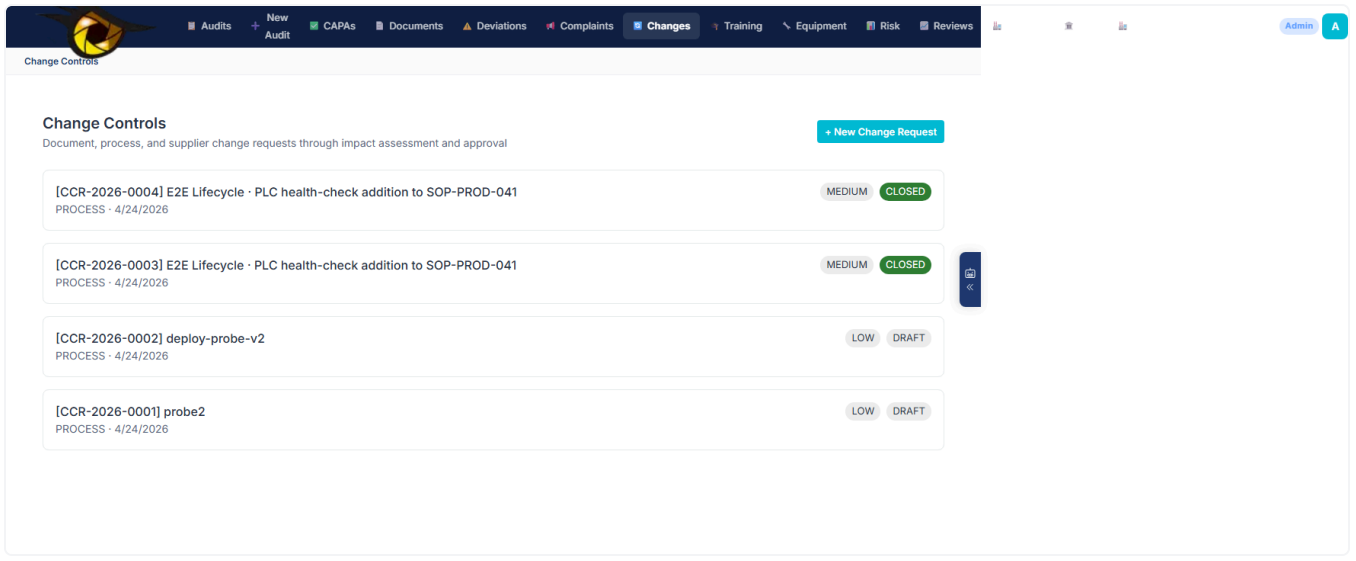
5. Full lifecycle walkthrough — one transaction, end-to-end

One change request walked from draft to closure with effectiveness verification.

#	Persona	State transition	Action	API	Observed	Result
1	Marcus Regulatory Affairs	– → DRAFT	+ New Change Request → fill PLC health-check addition / PROCESS / MEDIUM / Novexolimus / US → approvalSteps=[[1, QA Manager]]	POST /api/universal/change-controls	Row · changeNumber=CCR-2026-NNNN · status=DRAFT	PASS

Expected DB state after this step:

```
change_controls { _id, changeNumber: 'CCR-2026-NNNN', title, description, changeType: 'PROCESS', riskLevel: 'MEDIUM', status: 'DRAFT', approvalSteps: [{ stepOrder: 1, role: 'QA Manager', decision: 'PENDING' }] }
```



The screenshot shows the 'Change Controls' section of a software interface. At the top, there's a navigation bar with various icons and a 'Changes' tab selected. Below the navigation bar, the 'Change Controls' title is displayed, followed by a subtitle: 'Document, process, and supplier change requests through impact assessment and approval'. A '+ New Change Request' button is visible. The main area contains a list of four change requests, each with a title, a risk level (MEDIUM or LOW), and a status (CLOSED or DRAFT). The first two requests are 'CCR-2026-0004' and 'CCR-2026-0003', both with a risk level of MEDIUM and status of CLOSED. The next two are 'CCR-2026-0002' and 'CCR-2026-0001', both with a risk level of LOW and status of DRAFT. A 'state-screens/change-controls-list.png' caption is at the bottom.

2	Marcus Reg Affairs	DRAFT → SUBMITTED	PUT status=SUBMITTED	PUT /api/universal/change-controls/:id	status flips	PASS
3	Marcus Reg Affairs	SUBMITTED → SUBMITTED (with AI impact)	Click 'Classify regulatory impact' → AI returns US + EU routing	POST /api/ai/change-control/classify-impact	regulatoryImpact populated; classifier returned both routes	PASS
4	James Head of QA (approver)	SUBMITTED → APPROVED	Click 'Approve' on stepOrder=1 → APPROVED + comments → Submit	POST /api/universal/change-controls/:id/approval	All steps APPROVED → status=APPROVED	PASS

Expected DB state after this step:

```
change_controls.approvalSteps[0].decision = 'APPROVED'; status = 'APPROVED'
```

5	Marcus Reg Affairs	APPROVED → IMPLEMENTATION	PUT status=IMPLEMENTATION	PUT /api/universal/change-controls/:id	status flips	PASS
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#	Persona	State transition	Action	API	Observed	Result
6	Marcus Reg Affairs	IMPLEMENTATION → CLOSED	POST /verify- effectiveness with effective=true + verificationNotes	POST /api/universal/change- controls/:id/verify- effectiveness	status=CLOSED · effectivenessCheck recorded	PASS
Expected DB state after this step:						
change_controls { status: 'CLOSED', effectivenessCheck, effective: true }						

6. AI assistance map

Each AI assist is a separate endpoint. Outputs flow through the grounded-generation runtime (citation gate, confidence floor, schema validation, accept/reject audit trail).

AI agent	Attached to state(s)	Endpoint	Where in UI	What it returns	Provider
Regulatory Impact Classifier (Wave 2)	IMPACT_ASSESSMENT UNDER_REVIEW	POST /api/ai/change- control/classify- impact	Change detail · 'Classify regulatory impact'	Returns US (CBE-30 / PAS-NDA / Annual) + EU (Type IA/IB/II Variation) classification with reasoning	Free Gemini 2.5 Flash- Lite

7. Regulator-trace matrix

For each lifecycle state, the specific clauses + evidence Hawkeye captures so an inspector can audit the record. Maps to FDA / ISO / ICH / EU GMP source clauses.

State	Cited clauses	Evidence captured	Backing records
DRAFT	ICH Q10 §3.2.3	title + description + changeType + riskLevel	change_controls
IMPACT_ASSESSMENT	ICH Q10 §3.2.3(a)	regulatoryImpact + impactAssessment fields (auto- populated by AI classifier)	change_controls
APPROVED	21 CFR 211.100(b) 21 CFR Part 11	approvalSteps[] all APPROVED + e- sig per step	change_controls.approvalSteps + (electronic-signatures)
IMPLEMENTATION	ICH Q10 §3.2.3(b)	linkedDocumentIds (new SOP rev) + linkedTrainingIds + implementation notes	change_controls + document- controls (linked)
CLOSED	ICH Q10 §3.2.3(d)	verificationNotes + effectivenessCheck + effective=true + verifiedBy + verifiedAt	change_controls

8. Test results matrix

Suite	Scope	Result	Evidence
eqms-lifecycle.spec.ts · change-control	DRAFT → SUBMITTED → AI classify → APPROVED → IMPLEMENTATION → CLOSED	PASS	7/7 PASS · eqms-test-results-v2.pdf
Backend bug fix verification	Per-tenant changeNumber + compound unique index	PASS	v2.1 commit 56ddd85 + migration script ran

9. Known gaps + roadmap

- **Mandatory e-sig per approval step** — Wire /api/electronic-signatures/sign for each approve action. **HIGH**
- **Auto-link to document revision when changeType=DOCUMENTATION** — Today linkedDocumentIds is manual. Auto-create + link a new SOP rev. **MEDIUM**
- **Auto-trigger training when implementation completes** — If linkedDocumentIds includes a SOP that publishes, training auto-assign already fires. Confirm cross-link. **LOW**
- **AI suggested approver list** — Based on changeType + affectedProducts, suggest approvalSteps[] from a tenant policy table. **LOW**